



CY 242: Introduction to Molecular Spectroscopy

Course Code	CY 242
Course Name	Introduction to Molecular Spectroscopy
L-T-P-C	3-0-0-3
Prerequisites	Chemistry or Physics common course
Intended for	B. Tech. 3rd Year
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Quantum Mechanics**

Wave-particle duality, Schrodinger wave equation, Operators, Probability density, Matrix elements of operators and expectation values, One-dimensional problems in quantum mechanics - particle in a box, potential well, potential barrier and tunnelling. [12 Lectures]

- **Structure and bonding**

Hydrogen atom, Helium atom, Hydrogen molecule, Structure and Bonding, Nature of the chemical bond, Donor-Acceptor complexes, Charge transfer, Energy transfer, Conductance through DNA, Molecular electronic circuits, Single molecule transistors, Single molecule logic gates. [16 Lectures]

- **Spectroscopic Techniques**

Born-Oppenheimer Approximation, Molecular spectroscopy, Selection Rules, Vibrational and Rotational motion, Electronic Absorption and Emission Spectroscopy, Raman Spectroscopy. [14 Lectures].

Text and Reference Books

1. Jeanne L. McHale, **Molecular Spectroscopy**, Prentice Hall, 1998.
2. R. L. Liboff, **Introductory Quantum Mechanics**, Addison-Wesley, 2002.
3. Attila Szabo, **Modern Quantum Chemistry**, Dover, 2000.